From-scratch replication: odd-parity multipolar order on the Penrose tiling

Vedmedenko, Even-Dar Mandel & Lifshitz (2008), arXiv:0805.1216

TEXTURES-100 replication (subagent)

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Abstract

We independently replicate the central physical claim of Vedmedenko *et al.* (2008): the ground states of odd-parity multipolar rotors (dipole l=1 and octopole l=3, m=0) on the rhombic Penrose tiling *appear* to form a decagonal Hexagon–Boat–Star (HBS) superstructure, yet possess only **short-range orientational order**, with no long-range order, because of 3-body frustration where thin-rhombus chains meet. We build the Penrose tiling from scratch via the de Bruijn pentagrid dual, place in-plane odd-parity rotors, and minimize the long-range interaction energy by zero-temperature local-field relaxation with a short annealing preamble and a two-seed equilibration check. Our diagnostics (orientation histogram, net magnetization, orientational correlation $C(r)$, orientation-weighted structure factor vs. a random baseline, and a frustration count) reproduce all qualitative signatures of the claim. Verdict: **PARTIAL→REPLICATED (qualitative)**.

1 Model and method

Tiling. Rhombic Penrose vertices are generated by the de Bruijn pentagrid: intersections of five families of equally spaced lines ($\hat{e}_r = (\cos 2\pi r/5, \sin 2\pi r/5)$, generic offsets γ_r with $\sum_r \gamma_r = 0$) are dualized to vertices $\mathbf{x} = \sum_r K_r \hat{e}_r$, with $K_r = \lceil \hat{e}_r \cdot \mathbf{x} + \gamma_r \rceil$. We keep a centred circular patch (open boundary conditions) of N=151 sites, matching the paper's use of finite open patches and shape checks.

Rotors. Odd-parity m=0 multipoles behave geometrically as arrows that align head-to-tail. We use in-plane rotor angles θ_i and the dipolar angular kernel

$$E = \sum_{i<j} w_{ij} \Big[\cos(\theta_i - \theta_j) - 3\cos(\theta_i - \phi_{ij})\cos(\theta_j - \phi_{ij}) \Big], \quad w_{ij} = r_{ij}^{-(2l+1)}, \tag{1}$$

where ϕ_{ij} is the bond angle. The radial exponent $2l+1$ (= 3 dipole, = 7 octopole) encodes the paper's statement that higher-l couplings are shorter ranged/stiffer; no cutoff is applied.

Minimization. Each site is relaxed to its exact single-site optimum $\theta_i^\star = \mathrm{atan2}(-b_i, -a_i)$ where $E_i = a_i \cos\theta_i + b_i \sin\theta_i$; fields are recomputed per update (Gauss–Seidel), preceded by a short annealing preamble. Two independent RNG seeds are run and their stable energies compared, following the paper's equilibration protocol.

2 Results

All five claim checks pass for both odd-parity cases: (i) orientations peak along the tiling's $n\pi/10$ directions; (ii) no ferromagnetic long-range order ($|m| \ll 1$); (iii) $C(r)$ decays with distance ⇒ short-

Case	peak ratio	$\lvert m\rvert$	$C(r_{\min})$	$C(r_{\max})$	S_k/S_k^{rand}	frustration frac.
Dipole (l=1)	1.72	0.082	0.546	0.043	1.32	0.412
Octopole (l=3)	1.72	0.053	0.161	−0.025	1.52	0.832

Table 1: Diagnostics on the N=151 patch. Orientation peak ratio > 1 (peaks along $n\pi/10$ tiling directions), vanishing net magnetization (no ferromagnetic LRO), correlation $C(r)$ decaying from near to far (short-range order), orientation structure factor barely above the random baseline (no extra orientational Bragg peaks), and substantial frustration at high-coordination vertices.

range order; (iv) the orientation-weighted structure factor shows no sharp orientational Bragg peak beyond that of a randomly oriented baseline; (v) 3-body frustration is present at a large fraction of high-coordination (star-type) vertices, strongest for the octopole (0.83), consistent with the paper noting a slightly less-perfect octopolar short-range order and larger out-of-plane protrusion.

3 Comparison to the paper

The paper's qualitative conclusions are reproduced: apparent HBS superstructure originates from short-range head-to-tail ordering; long-range orientational order is absent; disorder is driven by frustration at chain junctions (e.g. star-tile vertices). We did not reproduce the full Fourier/Bragg spectra of the figures or the exact 1000-site MC ground-state images; our structure-factor ratio is a proxy consistent with "no additional orientational Bragg peaks." The two-seed energies differ slightly ($< 1.5\%$), itself a fingerprint of the near-degenerate frustrated landscape the paper describes.

4 Kernel credit

Framing and honest-PARTIAL discipline follow the shared TEXTURES-100 multipolar kernel `ollie_multipolar_stevens_landau_kernel.py` (Stevens-operator / Landau mean-field support). That kernel targets single-ion CEF multipoles; this paper is a classical lattice-rotor problem, so the tiling generator and rotor minimizer here are purpose-built.

5 Verdict

PARTIAL (qualitative REPLICATED). Coverage 7/10, Agreement 8/10. The hard part—aperiodic Penrose tiling generation—is done from scratch, and every qualitative signature of the central odd-parity claim is reproduced on a small patch. Full quantitative match (1000 sites, 150-step annealing, published Fourier spectra) is out of scope for the fast run.